



July 2026

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Upcoming

- July 17: STRUCTURES Jour Fixe: Olivier Pierre-Louis (Lyon University)
- July 22: CQD Colloquium: Sebastian Hofferberth (U. Bonn)
- July 24: STRUCTURES Day
- Aug 03-28: Les Houches School: Quantum Theory on All Scales
- Aug 25-28: YRC Schöntal Workshop
Register now! More details on page 2.

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Modified Gravity Theory Predicts Stable Non-Singular Black Holes and Gravastars

A recent study by STRUCTURES scientists Astrid Eichhorn and Pedro G. S. Fernandes presents a modified gravity theory that allows regular black hole solutions without mass-inflation instability, while also providing, for the first time, a gravastar as a solution of the equations of motion.

Singularities are a famous feature of General Relativity (GR). Nevertheless, a consistent description of black hole spacetimes is expected to yield regular geometries in which all physical quantities remain finite. An example of such a spacetime is the *Hayward metric*, a widely studied regular black hole solution with the asymptotic behaviour of a *Schwarzschild* black hole. However, geometries of this type are rarely derived directly from field equations, and when they are, they suffer from instabilities. In particular, regular black holes typically possess an inner horizon with non-zero surface gravity, where gravitational energy grows exponentially, leading to what is known as *mass-inflation instability*.

In the new work, the authors overcame both challenges by introducing a modified gravity theory that includes additional vector fields. As a result, additional degrees of freedom become what is known as *primary hair*: integration constants that modify the

spacetime geometry. The resulting geometries present a smooth core with finite curvature, a characteristic of Hayward-type solutions. Furthermore, by fixing one of the integration constants, the authors obtain *extremal* black holes, i.e., objects with vanishing surface gravity.

The most remarkable result of the study is the emergence of a *gravastar* solution directly from the dynamics of the theory. *Gravastars* are compact objects whose exterior matches that of a Schwarzschild black hole, while their core contains a dark-energy-like vacuum known as *de Sitter space*. This represents the first time that a gravastar-like object is a solution of a fully specified gravitational theory.

These results have important physical implications. Features previously viewed as “exotic” may arise from an extension of gravity without the need for “artificial” matter constructions. Additionally, since extremal black holes do not emit *Hawking radiation*, they could serve as dark matter candidates and may help avoid the infamous *information paradox*.

Original Publication:

Eichhorn, A., & Fernandes, P. G. (2026). Regular black holes without mass-inflation instability and gravastars from modified gravity. *Physical Review D*, 113(8), L081501.

SAVE THE DATE

Schöntal Discussion Workshop 2026

The STRUCTURES community is excited to announce the new edition of the 2026 Schöntal Discussion Workshop on *Dynamical Systems*. The event will take place from the 25th to the 28th of August at the quaint and charming Schöntal Abbey.

The aim of the workshop is to provide an opportunity for early-career scientists to gather and discuss topics that go beyond the standard mathematics and physics curricula, and to approach different subjects from their respective fields of research. The goal is to encourage interdisciplinary scientific exchange between the various areas of the cluster.

This year, the overarching topic of *Dynamical Systems* will be the protagonist. As a key concept in many physical, mathematical, computational and biological applications, we expect it to foster collaboration across different fields and spark



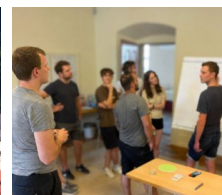
conversations that may lead to innovative and creative ideas and applications.

The workshop will guide participants through the fundamentals of dynamical systems, including the distinction between continuous and discrete time systems, core concepts such as stability, chaos, and bifurcation theories, and modern applications ranging from training algorithms to

Open for
MSc students,
PhD students &
postdocs



The workshop takes place every year in the tranquil and idyllic location of Schöntal Abbey.



Interactive discussions across disciplines inspire new perspectives. (Photo from a previous workshop).

recurrent neural networks (RNNs).

This workshop is funded by the Young Researchers Convent (YRC), a subgroup of STRUCTURES that brings together and supports early-career scientists.

The 2026 workshop is organized by Alicia Castro together with Sander Humerich, Luis Walter, and Alessandro Di Gregorio. This year's invited experts are Zahra Monfared and Tim Laux.

Image credit Schöntal Abbey: Wikipedia:Rosenzweig CC-BY-SA 3.0

MEMBER NEWS

STRUCTURES Professor Michela Mapelli Takes Leading Role in Europe's Next-Generation Gravitational-Wave Observatory

We are delighted to announce that STRUCTURES Professor Michela Mapelli has been appointed Chair of the Observation Science Board (OSB) of the *Einstein Telescope Collaboration*, one of the major international projects shaping the future of gravitational-wave astronomy.

The *Einstein Telescope* is the planned next-generation European gravitational-wave observatory. Designed to be about ten times more sensitive than current gravitational-wave detectors, it is expected to detect gravitational-wave events from across most of the observable Universe.

With more than 700 members, the Observation Science Board plays a central role within the collaboration: it coordinates the development of the scientific goals of the Einstein Telescope, including studies of compact objects, cosmology, multimes-

senger astronomy, data analysis, and synergies with other electromagnetic and gravitational-wave observatories.

Michela Mapelli has been a member of the Einstein Telescope Collaboration since its foundation in 2022 and has served as one of the lead editors of the "Blue Book". Her research focuses on understanding the formation and evolution of binary black holes and intermediate-mass black holes across cosmic time. Since 2023, Heidelberg University has hosted one of the main research units of the Einstein Telescope Collaboration in Germany, with more than 20 members spread across the Centre for Astronomy, the Institute for Theoretical Physics, the Max Planck Institute for Astronomy, and the Heidelberg Institute for Theoretical Studies.

"I am deeply honoured to serve as Chair



STRUCTURES Professor Michela Mapelli
Centre for Astronomy at Heidelberg University (ZAH) and Interdisciplinary Center for Scientific Computing (IWR)

of the Observation Science Board," says Mapelli. "The Einstein Telescope will open an entirely new window on the Universe, and I look forward to working with the international community and with my co-chairs, Archisman Ghosh (Ghent University) and Paolo Pani (La Sapienza University of Rome), to help shape its scientific vision."

COMMUNITY

We Are STRUCTURES

In our newsletter we regularly present short interviews with three randomly picked members of the Young Researchers Convent (YRC), a subgroup of STRUCTURES that brings together the early-career researchers of our scientific community. For this newsletter edition, we interviewed Sander Hummerich, Aina Nambena and Mrinal Sarkar:

Interview with Sander Hummerich:



Sander Hummerich
PhD student,
Bereau group

What are you working on?

I am developing probabilistic models that connect molecular simulations at different resolutions, primarily using concepts from generative machine learning. The goal is to

establish a probabilistic – essentially thermodynamic – bridge that enables high-resolution molecular insights from computationally cheaper low-resolution simulations.

What fascinates you about your research?

I am continually struck by the depth of the parallels between generative machine learning and statistical physics, and by how much each field stands to gain from the other. This perspective shapes my research on probabilistic models for multiscale molecular simulations, where generative ideas and thermodynamic structure interact.

What motivated you to enter science?

It was only through my MSc thesis that I came to appreciate how much I enjoy the scientific process, from engaging with the literature to formulating and testing my own ideas.

How does one recognize you?

You'll often see me biking from Philosophenweg to Weststadt or Kirchheim with my basketball bag. You might also spot me distributing flyers and posters for the Heidelberg Symposium.

Interview with Aina Nambena:



Aina Nambena
MSc student,
Bartelmann group

What are you working on?

I study how structures form in the Universe. In particular, I work on the kinetic theory of self-gravitating systems, using the Fokker-Planck equation to model collision terms that describe how gravitational interactions redistribute matter over time.

What fascinates you about your research?

In previous research projects, I worked on black holes and galaxies. Both are fascinating, but they somehow felt too small, too tangible. Now, my laboratory is the whole Universe – it can barely get any bigger or any more unimaginable – and I find that oddly poetic. I study how the Universe evolved from this homogeneous "soup" with tiny differences in temperature and density into the clumps we see today: galaxies, black holes, and all the cool stuff.

What motivated you to enter science?

When I was in school, the only female physicist I knew was Marie Curie, yet I could easily name a few dozen male scientists. Since physics is still a male-dominated field, I guess this is my small way of changing things.

How does one recognize you?

If you spot someone with curly hair playing card games in the second-floor kitchen of Philosophenweg 12, that's me.

Interview with Mrinal Sarkar:



Mrinal Sarkar
Postdoc,
Enss group

What are you working on?

I study the behaviour of complex systems at the intersection of statistical physics, nonlinear, and quantum dynamics – from synchronization and transport phenomena to phase

transitions in long-range disordered graphs. Beyond this, I collaborate with theorists and experimentalists on experimental synchrony, gene networks, and Physarum's computational abilities.

What fascinates you about your research?

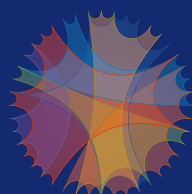
The joy of understanding and unraveling nature is unparalleled. After days or months of wrestling with ideas – whether through mathematics or experiment – there comes a surreal moment of clarity. The long struggle makes it worthwhile, and the new questions that follow only deepen the fascination.

What motivated you to enter science?

Well, it all started in my childhood, and I am grateful to my inspiring high-school teachers. Even before I fully realized it, I felt drawn to science because it seemed like the most logical way to address questions about everything. Over time, that curiosity grew into a passion.

How does one recognize you?

If you find a guy who is slightly messy-looking, with a lost look on his face while walking on the Philosophenweg, that's probably me.



The Young Researchers Convent (YRC) supports students and postdocs in realizing their own projects – from travel funding to the organization of seminars and talks. Any early-career researcher whose work fits within the concept of STRUCTURES can apply. If you are working in a STRUCTURES members' group or funded by STRUCTURES, you are directly eligible. In case of questions, feel free to reach out to: [structuresyrc\(at\)thphys.uni-heidelberg.de](mailto:structuresyrc(at)thphys.uni-heidelberg.de).



AWARDS & HONOURS

Teaching Prize for STRUCTURES Professor Richard Schmidt

We congratulate our member Richard Schmidt on receiving Heidelberg University's *Teaching Award* for his lecture *Theoretische Physik I*. The prize is awarded every semester by the Physics and Astronomy Department based on student feedback. Prof Schmidt received the award in the Winter Semester 2025/2026 for his first-semester lecture on classical mechanics. Students highlighted his commitment, engaging lectures, and his creativity in employing children's toys and household utensils to visualize the course

content. The result was a clear explanation of topics that are generally considered challenging during first-year physics courses. Some of the comments from the students read: "*Challenging, but beautifully explained.*" "*By far the best lecture I have attended so far.*"

Outside the classroom, Prof Schmidt leads the Quantum Matter Theory Group. His research focuses on the intersection of theoretical solid-state and atomic physics, where he studies complex quantum matter through theoretical few- and many-body



STRUCTURES Professor Richard Schmidt,
Institute for Theoretical Physics (ITP)

methods. He aims to discover novel states of quantum matter and identify ways for their experimental realization.

AWARDS & HONOURS

SAMOP Dissertation Award 2026 for Nikolas Liebster

We are happy to announce that STRUCTURES YRC alumnus Nikolas Liebster has received the 2026 *SAMOP Dissertation Award* from the *Atomic, Molecular, Quantum Optics and Photonics (SAMOP)* section of the *German Physical Society (DPG)*. The prize is awarded annually to recognize outstanding doctoral research as well as excellence in the written and oral presentation of a dissertation. Nikolas received the award for his dissertation *Pattern Formation and Supersolid Sound Modes in a Driven Superfluid*, completed in the research group of Markus Oberthaler at the Kirchhoff Institute for Physics (KIP) of Heidelberg University. The award was

presented at the DPG Spring Meeting 2026 in Mainz.

His research focuses on quantum systems far from equilibrium, where novel steady states can emerge that have different properties from the same system at equilibrium. In certain cases, theoretical frameworks from equilibrium physics can be applied to understand their behaviour.

In his thesis, Dr Liebster studied the emergence of such a steady state far from equilibrium, specifically in a *Bose-Einstein condensate* with periodically modulated interactions. The modulation results in the emergence of self-organized lattice structures, effectively crystallizing the system. Dr Liebster showed that this far-



Nikolas Liebster (YRC), now Postdoc at LMU.

from-equilibrium state has the properties of a supersolid, a strange state of quantum matter that is simultaneously solid and superfluid.

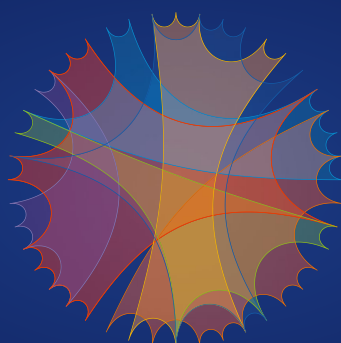
This work provides new insights into pattern formation and the nonlinear dynamics of quantum fluids.

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